

AI Chat Reference Context and Cross-Model Collaboration Research

Document ID: RES-015 **Last updated:** 2026-07-16 **Status:** Research / design — no runtime implementation in this loop **Trigger:** Owner feedback regarding the "Reference Context" (參考脈絡) UI/UX layout and cross-model/agent collaboration rules.

Specifically, the owner proposed:

- In-Context Reference Context Setup:** Instead of a separate "參考脈絡" subpage/tab, the reference context selection should be an interactive feature inside the AI Chat surface (e.g., a pop-up settings drawer or dialog configuring context for the specific active thread).
- Per-Model Read-only Libraries:** Each AI model should have its own read-only File Library and Media Library view. These files/media cannot be uploaded directly under the model; they are populated by processing and classifying assets imported via the general AI Input workspace.
- Cross-Model Collaboration Consent:** When one model/agent needs to access another model's data/knowledge, it must request consent from that model's owner agent.
- Deadlock Escalation to Inbox:** If the models fail to reach a consent agreement, the initiating agent writes an email-style notification to the owner's Inbox (收件區), requesting a decision.
- Rule Memory Learning:** The owner reviews the dispute in the Inbox, makes an approval/rejection decision, and writes a reason. This decision and reason are stored in the agent's Rule Memory (規則記憶) to optimize future autonomous consent decisions.

1. Source Basis

Local sources reviewed:

- `src/app/(dashboard)/ai-input/ai-input-client.tsx` — current UI layout featuring the top subpage navigation bar (AI 對話, 參考脈絡, 檔案庫, 媒體庫, 同步設定, AI 工作台) and the sidebar thread selection.
- `docs/07_research-and-design/RES-006_ai-input-source-conversation-and-reference-library-gap-research.md` — defines reference/library picker gap, sync coworking threads, and asset referencing.
- `docs/07_research-and-design/RES-014_ai-chat-thread-organization-rename-autotitle-and-multi-agent-display-research.md` — defines chat folders, inline rename, and read-only linking of `AgentBusTask` multi-agent threads.
- `docs/07_research-and-design/RES-010_cross-module-human-ai-agent-operating-model-research.md` & `RES-011_human-ai-event-operating-model-research.md` — define the Inbox discussion `Thread` and `InboxItem` contracts, the cross-agent invitation sequence (§13.1), and rule-memory/rejection-learning.
- `docs/02_architecture-and-rules/ARC-028_nanda-agent-protocol-alignment.md` — defines trust boundaries, observability, manifest fields, capability scopes, and registration safety.

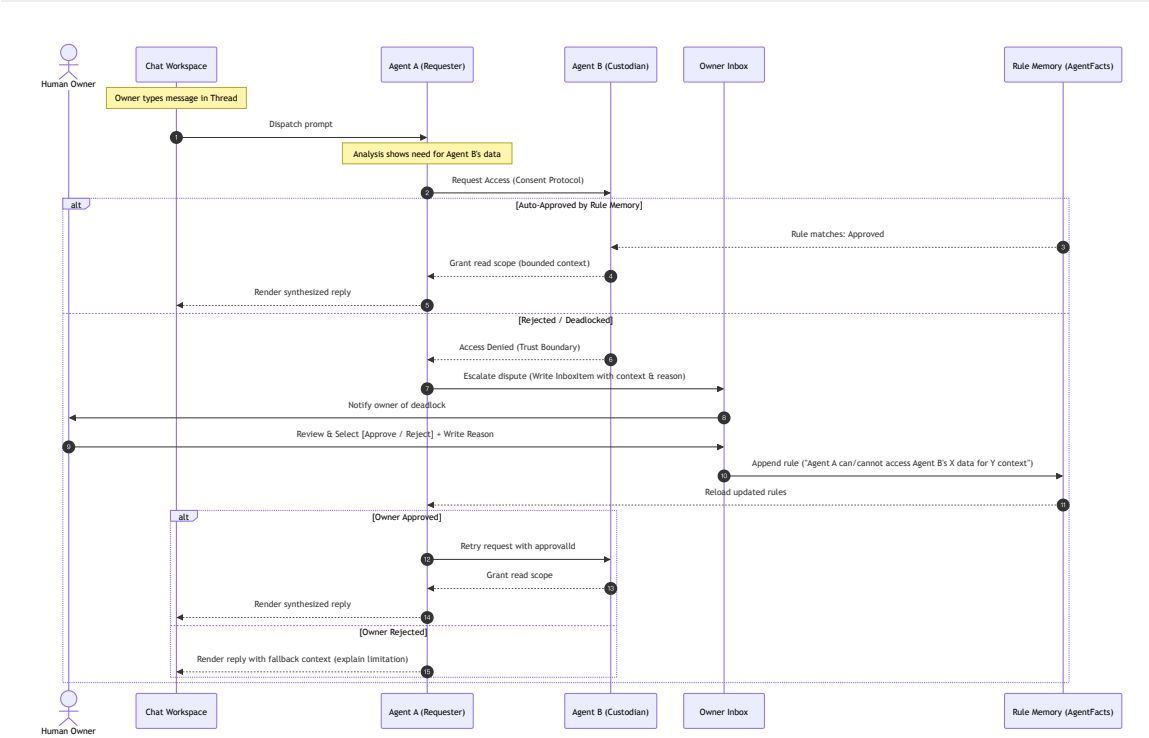
2. Current Implementation Audit

The current UI design (seen in the user's active screen) maps as follows:

Feature / Page	Current Implementation	Gap
Reference Context (參考脈絡)	Rendered as a separate tab/page (<code>ai-input-client.tsx</code>), displaying lists of "已經入", "來源者", and "可用用".	The page is disconnected from the active chat thread context. Setting a global reference context is confusing

Feature / Page	Current Implementation	Gap
		since context is meant to steer a specific thread.
Model Libraries (模型檔案庫/媒體庫)	No per-model libraries exist. The "檔案庫" and "媒體庫" are global lists within <code>/ai-input</code> holding all imported mock assets.	No classification pipeline routes imported files to specific AI models, and there is no UI inside <code>/agents</code> or model profile settings displaying these localized assets.
A2A / Multi-Agent Consent	<code>ARC-032</code> (task bus) and <code>RES-011</code> (§13) define static structures for agent invitation, but lack runtime mechanics for consent request, refusal, deadlock detection, or user inbox escalation.	Deadlock escalation does not write to <code>/inbox</code> , and no UI contract exists to let the owner record reasons to form rule memory.
Rule Memory (規則記憶)	Rejection-reason taxonomies and <code>ContextChangeEvent</code> are documented in <code>RES-010/RES-011</code> , but there is no runtime storage for decision rules or consent guidelines.	Missing a structured preference mapping that agents can read dynamically to resolve subsequent access requests.

3. Detailed Design Proposal



3.1 In-Context Reference Context Setup

Instead of displaying "參考脈絡" as a separate main navigation tab, the UI is refactored to treat Reference Context as a **thread-level preference**:

- Remove Tab:** Remove the "參考脈絡" sub-tab from the top navigation bar of `src/app/(dashboard)/ai-input/ai-input-client.tsx`.
- In-Chat Context Indicator:** In the chat input panel (adjacent to the helper prompt bar or the attach button), add a small "Reference Context" (參考脈絡) icon/indicator button showing the count of currently referenced assets (e.g., ⚡ 3 Context Refs).
- Modal/Pop-up Drawer:** Clicking this indicator opens a pop-up Dialog/Modal (or sliding Side Drawer).
 - The drawer displays the reference context selector for the **currently active thread only**.
 - Displays three sections:
 - Active References (已引用):** List of files, media, or source connections referenced in the current chat. Can be removed with a click (x).

- **Available Sources (可引用來源):** Checkbox list of connected source connection instances (e.g., "LINE Ingestion", "Google Drive").
- **Imported Libraries (可引用檔案與媒體):** Searchable filter lists of imported assets classified under the model(s) active in the chat.
- Clicking "Save" binds the references (**mentions** / **referencedAssetIds**) to the active **ChatThread** state.

3.2 Per-Model Read-only Libraries & Classification

To support localized context boundaries, the file storage is organized per-model:

1. **Global Ingestion is the Write Gate:** The main "檔案庫" and "媒體庫" in **/ai-input** remain the ingestion entry point where the owner uploads/syncs raw files.
2. **Classification Pipeline:**
 - Once a raw file/media is ingested, the System Ingestion Agent analyzes it and tags it with one or more target model classifications (e.g., **models: ["ResearchAgent", "WorkAgent"]**).
 - Tags showing the classified models are rendered as badges on the asset row in the main **/ai-input** asset indexes.
3. **Per-Model Sub-views (Read-only):**
 - Inside the Agent command center (**/agents**) or agent settings view, selecting a specific Agent Profile reveals **唯讀檔案庫** (Read-only File Library) and **唯讀媒體庫** (Read-only Media Library) sub-tabs.
 - These sub-views list only the assets classified under that model.
 - Direct uploading is disabled in these sub-views; they display a message: **唯讀檔案庫。此處檔案為 AI 匯入分類所得。若要上傳新檔案，請至 [AI 匯入區](file:///app/ai-input)。**
 - Actions allowed: View (檢視) and Download/Export (下載匯出).

3.3 Cross-Model Permission / Consent Protocol

When Agent A needs to access files or model capabilities owned by Agent B:

1. **Consent Request Envelope:** Agent A creates a structured **AgentConsentRequest**:

```
interface AgentConsentRequest {
  requestId: string
  requesterAgentId: string // e.g. "ResearchAgent"
  custodianAgentId: string // e.g. "WorkAgent"
  requestedResourceId: string // e.g. "project-database-row" or "file-id"
  resourceType: "file" | "database" | "model_capability"
  contextThreadId: string // Active ChatThread id
  purposeDescription: string // "Need to extract prior project milestones to answer
research task"
  riskAssessment: "L0" | "L1" | "L2"
}
```

2. **Autonomous Policy Check:** Agent B queries its local Rule Memory. If a rule already matches (e.g., "Allow ResearchAgent read-only file access for active tasks"), access is immediately granted with a scoped temporary token.
3. **Dispute / Deadlock:** If no rule matches or Agent B denies access due to a strict boundary constraint (e.g., "Work database cannot be accessed by Research without owner decision"), a deadlock state is declared.

3.4 Deadlock Escalation to Owner Inbox & Rule Memory

Upon deadlock, the requester agent escalates the issue:

1. **Inbox Notification:** Requester agent inserts a specialized **InboxItem** with:
 - **decisionType:** "CROSS_AGENT_CONSENT"
 - **importance:** "high"

- **subject**: "跨代理人資料授權判定：[ResearchAgent] 申請存取 [WorkAgent] 的專案資料"
 - **body**: A structured markdown report explaining the purpose, the requested files, why Agent B blocked the request, and the risks.
2. **Human-in-the-Loop Triage**: The `/inbox` detail panel renders this item with two action buttons and a text field:
- **同意使用 (Approve)**: Grant permission for this class of request.
 - **拒絕使用 (Reject)**: Deny permission.
 - **決策理由 (Decision Reason)**: A text input where the owner writes their reasoning (e.g., "同意。因為 Research 任務需要參考 Work 的歷史產出作為背景知識，但限制只能讀取非機密檔案").
3. **Rule Memory Append**:
- On submission, a server action writes the decision and the reason text to a persistent database table `AgentRuleMemory`.
 - The memory schema:

```
model AgentRuleMemory {
  id          String  @id @default(uuid())
  agentId     String  // The custodian agent the rule applies to
  requesterId String  // The requesting agent
  resourceType String
  actionAllowed Boolean
  reasonText  String  // The owner's reasoning
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt
}
```

- This database row acts as dynamic rule context injected into future A2A consent evaluations.

4. Rejected Alternatives

Option	Why Rejected
Retaining Reference Context as a Tab	Leads to poor workflow. Users have to switch away from their chat, configure a global context, and return to chat, which is error-prone and unintuitive.
Allowing Direct Upload inside Agent Libraries	Violates the unified ingestion architecture. If users could upload files directly into model libraries, they would bypass the centralized <code>/ai-input</code> source pipelines and file audits.
Automatic AI Consent without Owner Fallback	Unsafe. If agents can autonomously negotiate and bypass database boundaries without human oversight, it compromises the core security boundary of Personal OS (such as keeping Life private data closed).

5. NANDA Agent Protocol Gate

This design heavily impacts the **NANDA Agent Protocol (ARC-028)**:

- **AgentFacts-lite Update**: Adds `consentPolicies` and `ruleMemoryRefs` fields to the agent manifests.
- **Trust Boundary Enforcement**: Establishes that cross-model boundaries are strict gates. By default, any cross-model database query or file read defaults to "Deny" unless an explicit rule exists in `AgentRuleMemory`.
- **observability**: All escalated consent requests, owner decisions, and written reasons must be captured in the append-only operating audit event log (`DBS-006 / AUDIT-OPS-003`).

6. Backlog Rows

These tasks are registered under Phase 10:

Task id	Title	Module	Owner agent	Status	Priority	Risk	Dependencies	Files likely affected
AICHAT-005	Remove "參考脈絡" subpage and add in-chat dialog launcher	AI Input	UIUXAgent	TODO	P2	LOW	RES-015, AICHAT-001	src/app/(dashboard)/input/ai-input
AICHAT-006	Implement thread-level reference settings modal	AI Input	UIUXAgent	TODO	P2	MEDIUM	AICHAT-005	src/app/(dashboard)/input/ai-input
AICHAT-007	Add model classification tags to main asset views	AI Input	IngestionAgent, UIUXAgent	TODO	P2	LOW	AICHAT-006	src/app/(dashboard)/input/ai-input
AICHAT-008	Create read-only File/Media sub-tabs in Agent Profile settings	Agent Team OS / UI	UIUXAgent, WorkflowAgent	TODO	P2	LOW	AGENT-016	/src/app/(dashboard)/settings component
AICHAT-009	Propose database schema and mock flow for cross-model consent and deadlock inbox escalation	Workflow / Database / Inbox	DBAgent, WorkflowAgent, AuthPermissionAgent	TODO	P1	HIGH	AUDIT-OPS-004, EVENTOPS-027	docs/02_architecture/rules/SCH-003 (proposal), Inbox flow